

Time Series & Correlation Assignment

Question 1

Why identical correlation coefficients do not imply the same relationship: Pearson's correlation coefficient (r) measures only the strength and direction of a linear relationship between two variables.

- **Pair A:** reflects a true, consistent linear dependence.
- **Pair B:** has a non-linear (curved) relationship where Pearson correlation fails to capture the true underlying structure. Pearson's r is also extremely sensitive to outliers; a single extreme observation can artificially inflate or deflate the correlation coefficient, similar to Anscombe's Quartet.

What an analyst should examine:

- **Scatter plots:** visually inspect the data for non-linearity, heteroscedasticity, or clustering.
- **Outliers and leverage points:** identify extreme values that disproportionately influence the summary statistic.
- **Alternative metrics:** use rank-based non-parametric correlations such as Spearman's rank or Kendall's tau if appropriate.

Question 2

What high lag-1 autocorrelation actually tells us: High lag-1 autocorrelation in closing prices indicates strong persistence or inertia in price levels, meaning today's price is very close to yesterday's price.

Why it does not imply predictable or profitable returns:

- **Prices vs. returns:** High autocorrelation in prices is a standard feature of non-stationary series such as a random walk.

$$S_t = S_{t-1} + \epsilon_t$$

Trading profitability depends on predicting returns, not price levels:

$$R_t = (S_t - S_{t-1}) / S_{t-1}$$

In a random walk, price levels can be highly autocorrelated while price changes are unpredictable white noise.

- **Transaction costs and spreads:** even if a small predictable component exists in returns, bid-ask spreads and execution costs can eliminate the theoretical profit margin.

Question 3

Why this is a problem: An upward trend implies non-stationarity; specifically, the mean of the series changes over time. Models trained on non-stationary data can break down out of sample because statistical properties are not stable across time.

Transformations to achieve stationarity:

- **Differencing:** take first differences.

$$\Delta S_t = S_t - S_{t-1}$$

- **Log returns:** calculate changes in log prices, which remove the price-level trend and are often more suitable for modeling.

$$r_t = \ln(S_t) - \ln(S_{t-1})$$

Question 4

Diagnostic advantage of keeping provider datasets separate: Keeping Bloomberg and FactSet data separate creates a clean framework for cross-provider benchmarking and validation.

What differences help you learn:

- **Data handling and adjustments:** discrepancies can reveal different treatments of stock splits, dividends, spinoffs, and other corporate actions.
- **Survivorship and restatement bias:** differences can reveal how providers handle delisted firms, historical revisions, or point-in-time filing dates.
- **Strategy sensitivity:** comparison helps determine whether a quantitative signal is robust or overly sensitive to vendor-specific formatting and timing definitions.

Question 5

Proof:

1. Express $X_t - X_{t-1}$ using the log definition:

$$X_t - X_{t-1} = \ln(S_t / S_0) - \ln(S_{t-1} / S_0)$$

2. Apply the logarithm quotient rule:

$$X_t - X_{t-1} = \ln[(S_t / S_0) / (S_{t-1} / S_0)] = \ln(S_t / S_{t-1})$$

3. Express S_t / S_{t-1} in terms of nominal return R_t :

$$1 + R_t = S_t / S_{t-1} \Rightarrow S_t / S_{t-1} = 1 + R_t$$

4. Substitute back:

$$X_t - X_{t-1} = \ln(1 + R_t)$$

Explanation of the approximation: Using the Taylor series expansion of $\ln(1+r)$ around $r=0$:

$$\ln(1+r) = r - r^2/2 + r^3/3 - \dots$$

When $|r|$ is small, as with many daily asset returns, the higher-order terms become negligible. Therefore:

$$\ln(1 + R_t) \approx R_t$$

which yields:

$$R_t \approx X_t - X_{t-1}$$